

Artificial Intelligence for Financial Services

Technology, Data, and Market Dynamics in Modern Investing

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Finance at MIT
Where ingenuity drives results

Overview

- Technology and data
 - New opportunities with AI tech and resources
 - Scale **everything**
- Investment strategies
 - Algorithmic trading past and present
- Adaptive investment strategies for adaptive markets
 - Reinforcement learning and financial markets
 - Learning to trade (profitably!) using RL
 - Trading to learn: RL as a source for financial insights

Cutting edge has shifted from prediction to adaptation

- In modern markets, the investment question is increasingly shifting from
 - “What can we **predict**?” to
 - “When, how – and why – should the system **act**, **pause**, **scale**, or **learn**?”
- Tools can be copied. Adaptive learning systems, less so.
 - When many investors get better tools, the market changes
 - The same tool that **finds** an opportunity can **erase** it when widely adopted

Traditional NLP Pipeline

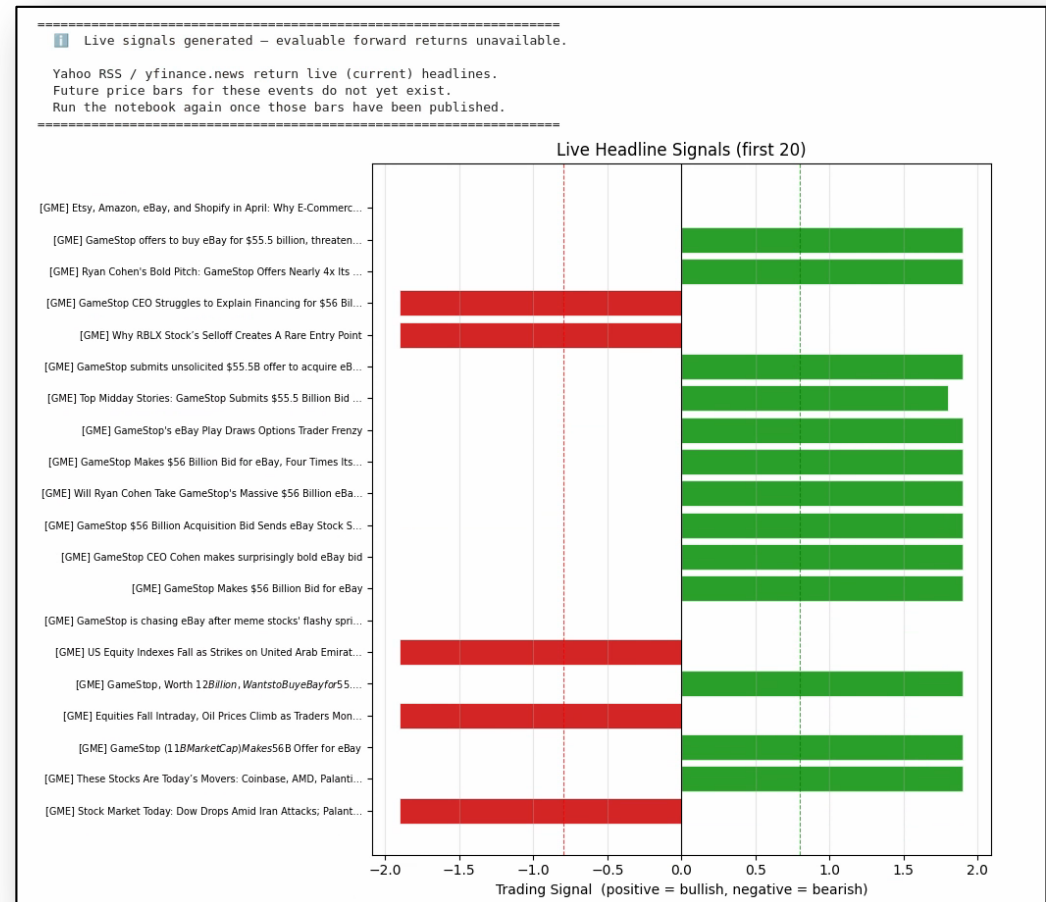
- Define universe/watchlist
- Define channels and signals
 - Scan feeds and streams
 - Identify candidate events
 - Count event arrivals and rate
 - Raise keyword flags
 - Compute sentiment scores
 - Classify and categorize documents/tweets/articles
 - Identify impacted firms and securities
- Analyze events and take action
 - Buy, sell, hold
 - Open, close, or hedge existing positions
 - Multiple horizons and frequencies



Source: Bloomberg

Adding LLMs to the mix

- Large language models can compress multiple elements of the pipeline
 - Text tokenization
 - Classification
 - Prediction
 - Decision
 - Agent-based actions
- Advantages
 - Efficiency, consistency
- Drawbacks
 - Loss of control, selection, transparency, reproducibility



LLMs for LLMs

- Large language models also aid in the design and implementation
 - Jupyter notebook written by Claude Opus 4.6, based on
 - Instructions from ChatGPT 5.4
 - Execution on Nvidia DGX Spark



- Advantages of local execution
 - Efficiency, consistency
 - Security, QoS

```

demo-news-prompts.txt x financial_news_trading_signals.ipynb x
1
2 Create a fully runnable Jupyter notebook for a finance/NLP classroom demo titled:
3
4 "From Financial News to Trading Signals: Event Extraction with Local LLMs"
5
6 The notebook should be designed for an MBA / graduate fintech machine-learning course. Its
  goals are:
7
8 1. Pull real market price data.
9 2. Pull real financial news headlines.
10 3. Use a local LLM to extract structured financial events from headlines.
11 4. Convert extracted events into simple trading signals.
12 5. Match events to price data.
13 6. Compute forward returns over configurable horizons.
14 7. Clearly distinguish between live signal generation and evaluable historical event-study
   results.
15 8. Be robust across mixed hardware, including GPU workstations and student laptops.
16
17 Important design requirements:
18
19 - The notebook must be self-contained and runnable.
20 - It should not require an API key for the LLM.
21 - It should support two local LLM backends:
22   - Hugging Face Transformers
23   - Ollama
24 - It should default to auto-selecting Hugging Face when a working GPU is available, and
   otherwise use Ollama.
25 - It should also allow explicit backend selection with a parameter such as:
26   - LLM_BACKEND = "auto"
27   - LLM_BACKEND = "hf"
28   - LLM_BACKEND = "ollama"
29
30 Market data requirements:
31
32 - Use yfinance to pull adjusted daily close prices from Yahoo Finance.
33 - Include a configurable ticker list, for example:
34   - TICKERS = ["AAPL", "MSFT", "TSLA", "XOM"]
  
```

Algorithmic trading: keeping score

- Traditional approach
 - **Describe:** model the markets and environment
 - **Predict:** forecast prices, returns, other features, etc.
 - **Decide:** determine trades, timing, quantities, hedges
- Predictions, signals, forecasts, and trades
 - Statistical score keeping
 - Economic score keeping
- Modern approach:
 - Focus on **outcomes** and **objectives**
 - Solve for **actions**, not predictions

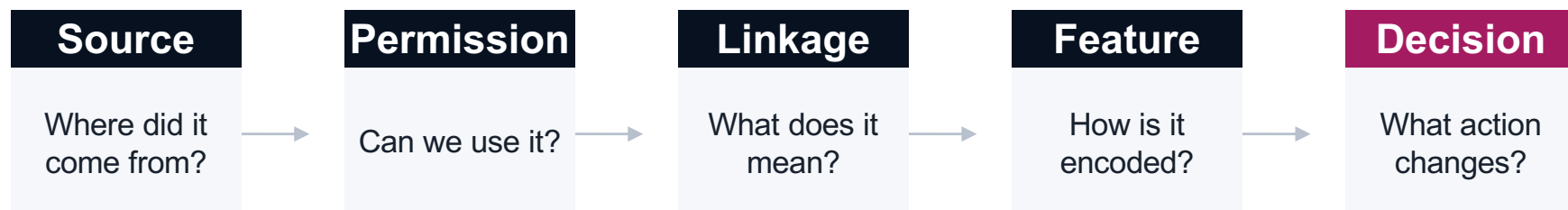
Learning loops provide precision, control, improvement

- Elements of quantitative investing
 - Data, new & used, structured & unstructured, frequencies high & low
 - Models and tools
 - Selection of universe, securities, portfolios
 - Feedback, monitoring, measurement, compliance
 - Learning and adaptation



The right data enables learning the right features

- Data is valuable when it is relevant and timely
 - Features are more critical than models, yet...
 - Feature relevance can evolve slowly or shift suddenly



Markets punish static prediction

- Profitable, scalable, successful algorithms can fail
 - Competition, crowding, feedback, market impact
 - Capacity, scale, and transaction costs
 - Regime shifts
 - Non-stationarity
 - Implementation failures
 - Fragile modeling framework
- Even when signals have been predictive,
 - Will they survive while the market is changing?
 - How can you tell?

Learning to trade with reinforcement learning

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What is reinforcement learning

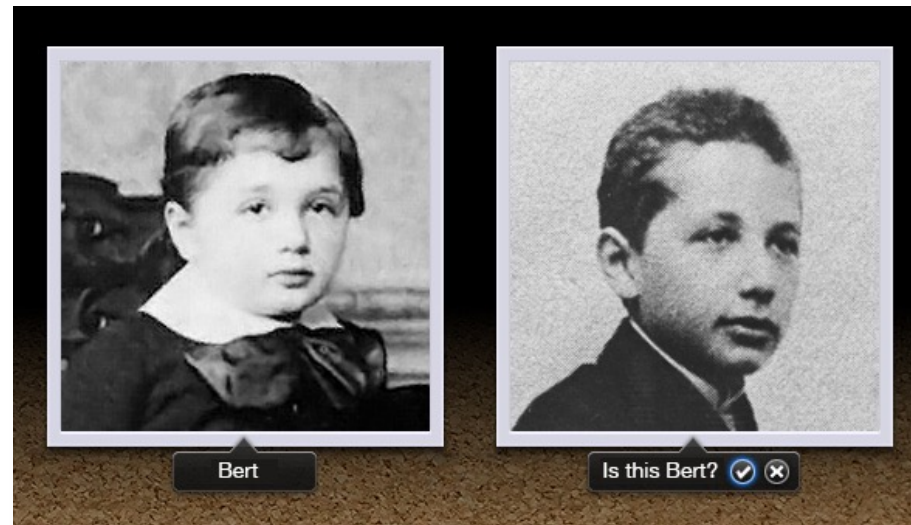
- Reinforcement learning is a technique for **learning new skills** through experience
 - Start in a new, unknown environment
 - Try out **everything** possible
 - Watch what happens
 - Actions that lead to good outcomes: do more often
 - Actions that lead to bad outcomes: do less often
 - Keep mixing things up!

What is reinforcement learning

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 - Keep mixing things up!
- Aspects that are surprisingly helpful in financial markets:
 - It's **slow**
 - It's **tentative**
 - It's **uninformed**
 - It's **adaptive**
 - It's [interpretable](#)

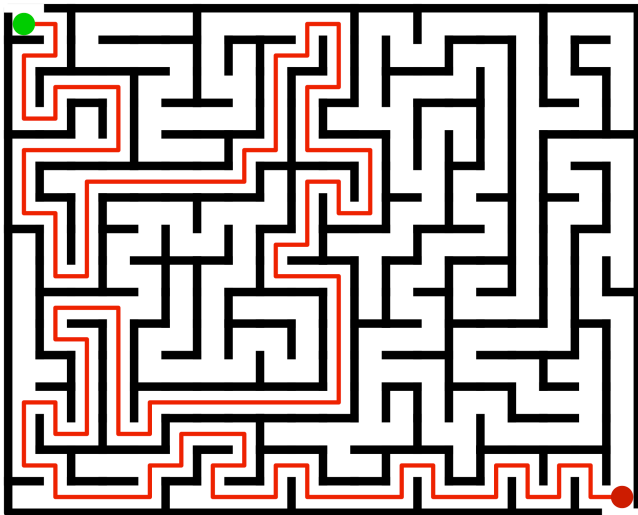
Dimensions and attributes of machine learning

- **Supervised vs. unsupervised**
- Offline vs. online
- Environment model vs. discovery
- Stationary vs. non-stationary
- Exploitation vs. exploration
- Loss functions vs. goal-seeking
- Interpretability



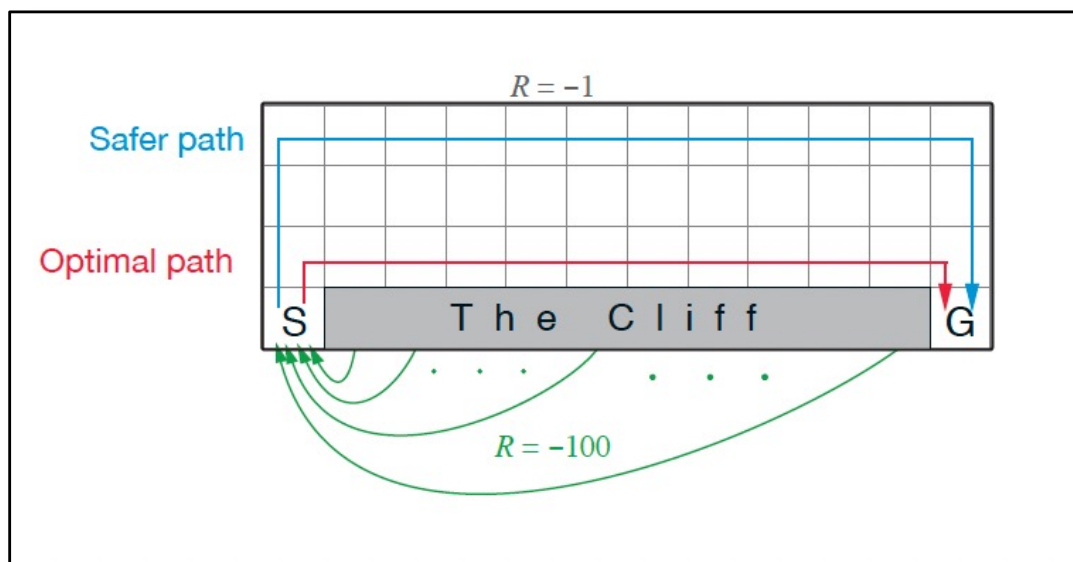
Reinforcement learning

- **Reinforcement learning** draws on language of **behavior** and analytics of finance.



Reinforcement learning

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Source: Sutton & Barto

Reinforcement learning and financial markets

- **Reinforcement learning** draws on language of **behavior** and analytics of finance.
- Let's apply it to a classic problem: **learning to trade**.
 - Only inputs: market data and history
 - Only actions: buy, sell, or hold
 - Only goal: maximize long-term return
- Lessons about finance and about trading
- Tool for exploring market structure, innovations, crises, phase transitions, etc.
- Applications and extensions

Concepts for reinforcement learning

- States
- Actions
- Rewards
- Goal
- Values
- Performance

Definitions and notation:

- States $s \in S_t$: states available at time t .
- Actions $a \in \mathcal{A}(S_t)$: actions available from states at time t .
- Transition probabilities $p(s'|s, a)$: probability to end in state s' after taking action a from initial state s .
- Rewards $r(s, a, s')$: reward (distribution) resulting from taking action a from initial state s leading to successor state s' .
- Policy $\pi(a|s)$: probability (distribution) for action a to be selected, given initial state s .
- Value function $V_\pi(s)$: value of being in state s under policy π .
- State-action value function $q_\pi(s, a)$: value of taking action a under policy π from a given state s .

Bandits and beyond

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Setup for reinforcement learning

- Both the **policy** (choice of action) and the **outcomes** (reward as well as subsequent state) may be **stochastic**
- Simplified problem: “**N-armed bandit**”
 - Only a single state
 - Choice of action and subsequent reward therefore **do not depend** on initial state or determine subsequent state.



Exploit or explore?

- A **greedy** (local) policy selects the action which is believed to have the highest value
- Problems:
 - Is the value **estimated** accurately?
 - Is a locally optimal policy also **globally optimal**?
- An **ϵ -greedy** policy selects:
 - With probability ϵ , an action **at random**
 - With probability $1-\epsilon$, follows greedy policy

A simple bandit algorithm

Initialize, for $a = 1$ to k :

$Q(a) \leftarrow 0$

$N(a) \leftarrow 0$

Repeat forever:

$A \leftarrow \begin{cases} \arg \max_a Q(a) & \text{with probability } 1 - \epsilon \quad (\text{breaking ties randomly}) \\ \text{a random action} & \text{with probability } \epsilon \end{cases}$

$R \leftarrow \text{bandit}(A)$

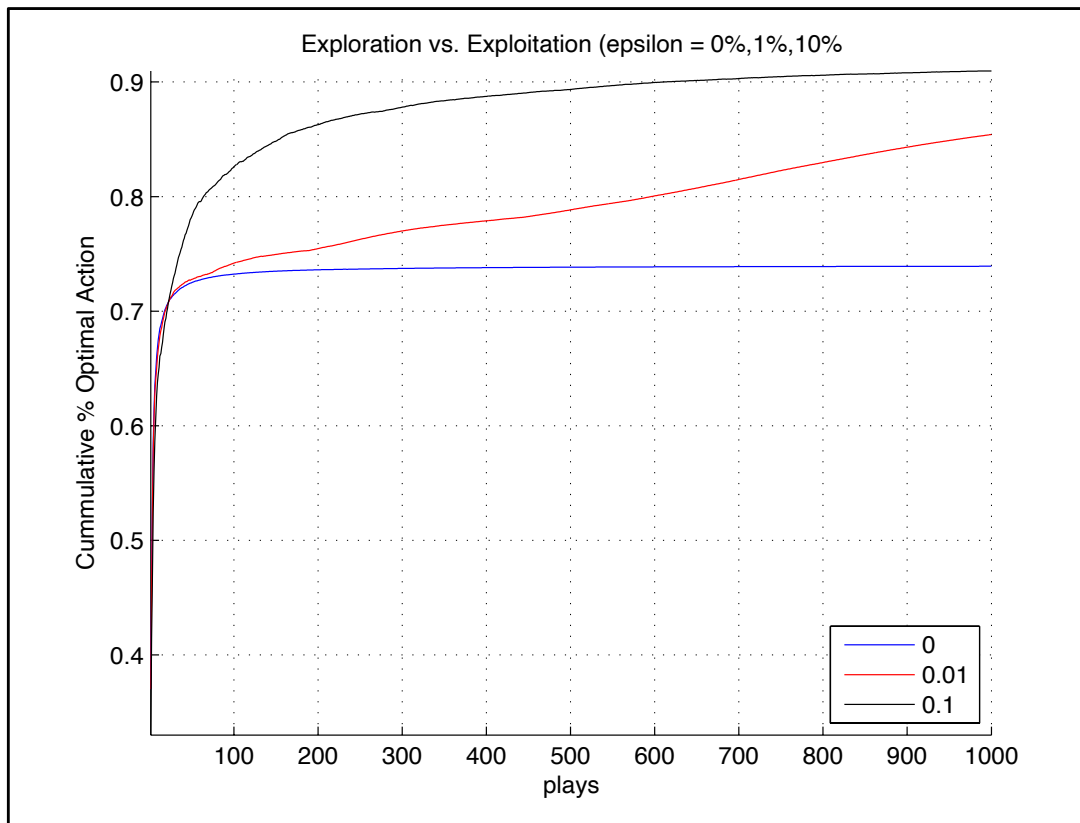
$N(A) \leftarrow N(A) + 1$

$Q(A) \leftarrow Q(A) + \frac{1}{N(A)} [R - Q(A)]$

Source: Sutton & Barto

$$\begin{aligned} Q_{t+1}(s, a) &= Q_t(s, a) + \alpha [R_t - Q_t] \\ &= (1 - \alpha)Q_t(s, a) + \alpha R_t \end{aligned}$$

Continual exploration leads to better long-term policies



Learning to trade

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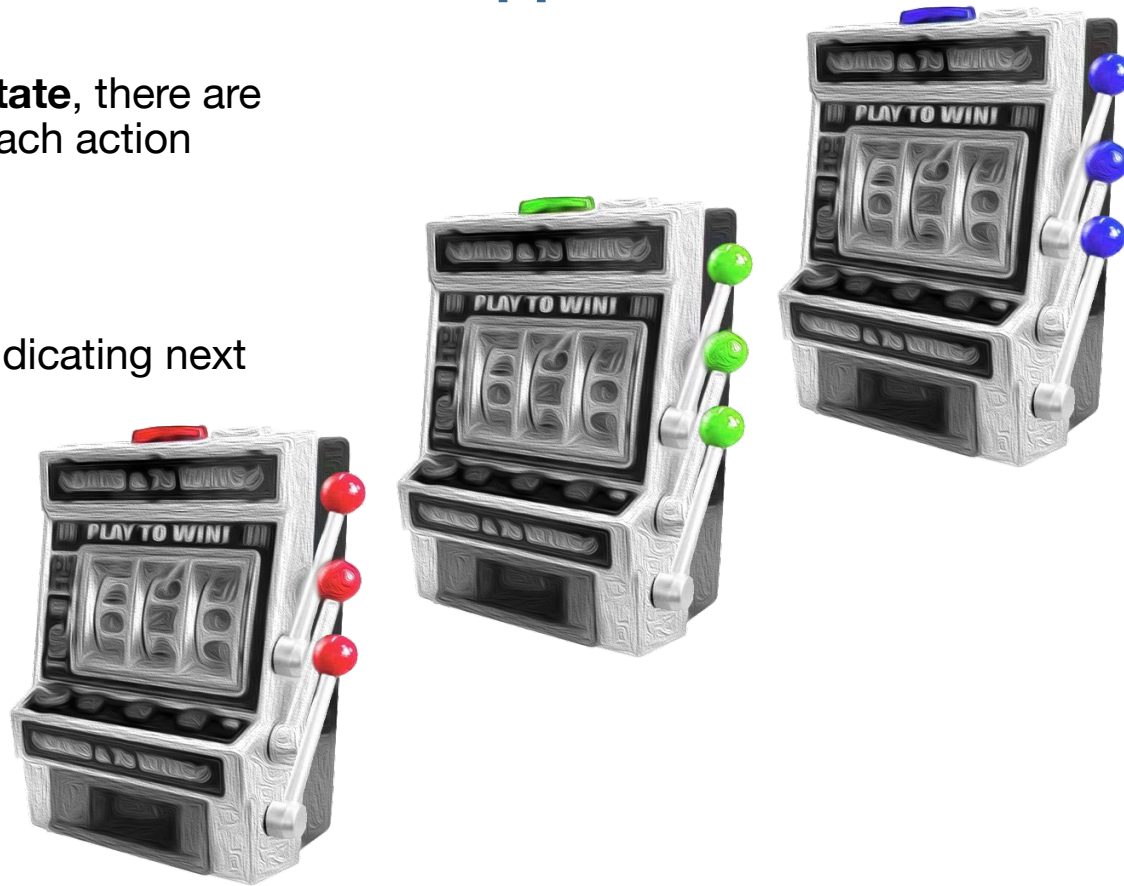
Act now! RL is where the action is

- We consider a finite set of discrete actions:
 - Buy
 - Sell
 - Hold
- Equivalently, we can cast in terms of actions required to arrive at specified holdings given current holdings, e.g.,
 - Long
 - Short
 - Flat
- Generalizations: trade any quantity at any time



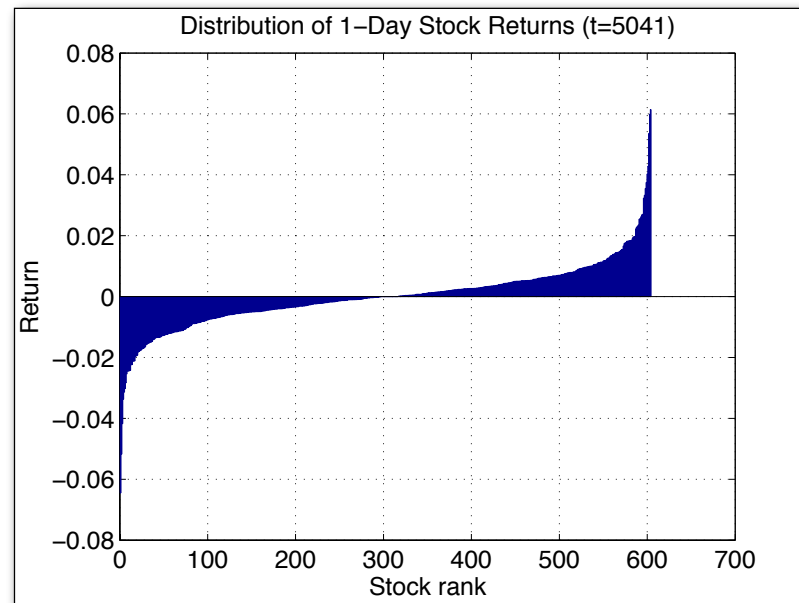
Different states offer different opportunities

- For each identified **state**, there are **distinct values** for each action
- Choose a lever and
 - Receive a **reward**
 - Receive a color, indicating next machine to use



States depend on the market and the portfolio

- For each identified **state**, there are distinct values for each action
- State variables (external):
 - Market data set
 - Derived signals (fundamentals, macro, news, tweets, etc.)
 - Example: winners vs. losers, **deciles of excess return**.
- State variables (internal):
 - Portfolio holdings



The challenge: “efficient” markets behave randomly!

Proof That Properly Anticipated Prices Fluctuate Randomly

Paul A. Samuelson
Massachusetts Institute
of Technology

The Enigma Posed

"In competitive markets there is a buyer for every seller. If one could be sure that a price will rise, it would have already risen." Arguments like this are used to deduce that competitive prices must display price changes over time, $X_{t+1} - X_t$, that perform a random walk with no predictable bias.

Is this a correct fact about well-organized wheat or other commodity markets? About stock exchange prices for equity shares? About futures markets for wheat or other commodities, as contrasted to the movement of actual "spot prices" for the concrete commodity.

Or is it merely an interesting (refutable) hypothesis about actual markets that can somehow be put to empirical testing?

Or is it a valid deduction (like the Pythagorean Theorem applicable to Euclidean triangles) whose truth is as immutable as

wheat prices generally rise (presumably because of storage costs) from the July harvest time to the following spring and drop during June? Is the fact that the price of next July's future shows much less strong seasonal patterns a confirmation of the alleged truism? If so, what about the alleged Keynes-Hicks-Houthakker-Cootner pattern of "normal backwardation," in which next July's wheat future could be expected to rise in price a little from July harvest to, say, the following March (as a result of need of holders of the crop to coax out, at a cost, risk-disliking speculators with whom to make short-hedging transactions); and what about the Cootner pattern in which, once wheat stocks become low in March, processors wishing to be sure of having a minimum of wheat to process, seek short-selling speculators with whom to make long-hedging transactions, even at the cost of leaving the take-out position dropping a

Whatever the answers to these questions, I think we can suspect that there is no *a priori* necessity for actual Board of Trade grain prices to act in accordance with specific probability models. Perhaps it is a lucky accident, a boon from Mother Nature so to speak, that so many actual price time series do behave like uncorrelated or quasi-random walks. Thus, Maurice Kendall almost proves too much when he finds negligible serial correlation in spot grain prices. For reasons that I shall discuss, we would not be too surprised to find this property in futures price changes. But surely spot prices ought to vary with shifts in such supply and demand factors as weather, crop yields, and crop plantings; or population, income, and taste changes. Who says that the weather must itself display no serial correlation? A dry month does tend to be followed by a dryer-than-average month because of persistence of pressure patterns, etc. Perhaps it is true that prices depend on a summation of so many small and somewhat independent sources of variation that the result is like a random walk. But there is no necessity for this. And the fact, if it is one, is not particularly related to perfect competition or market anticipations. For consider a monopolist who sells (or buys) at fixed price. If the demand (or supply) curve he faces is the resultant of numerous independent, additive sources of variation each of which is limited or small, his resulting quantity $\{q_t\}$ may well behave like a random walk, showing variations like the normal curve of error.

At this point the reader may object that a stock price systematically overstates the

You never get something for nothing. From a nonempirical base of axioms you never get empirical results. Deductive analysis cannot determine whether the empirical properties of the stochastic model I posit come at all close to resembling the empirical determinants of today's real-world markets. That question I shall not here investigate. I shall be content if, for once, I find definite and unambiguous content to the arguments of the opening paragraph—arguments which have long haunted economists' discussions of competitive markets.

A General Stochastic Model of Price

Let $\{X_1, X_2, X_3, \dots, X_t, X_{t+1}, \dots, X_{t+T}, \dots\}$ represent the time sequence of prices—for example the price of spot #2 wheat in Chicago (or it could be a vector of prices or even quantities of several different goods). Given knowledge of today's price and of past prices $\{X_1, X_2, \dots\}$, suppose we cannot know with certainty tomorrow's price X_{t+1} , or any future price X_{t+T} . Suppose there is at best a probability distribution for any future price, whose form depends solely on the number of periods ahead over which we are trying to forecast prices, given by

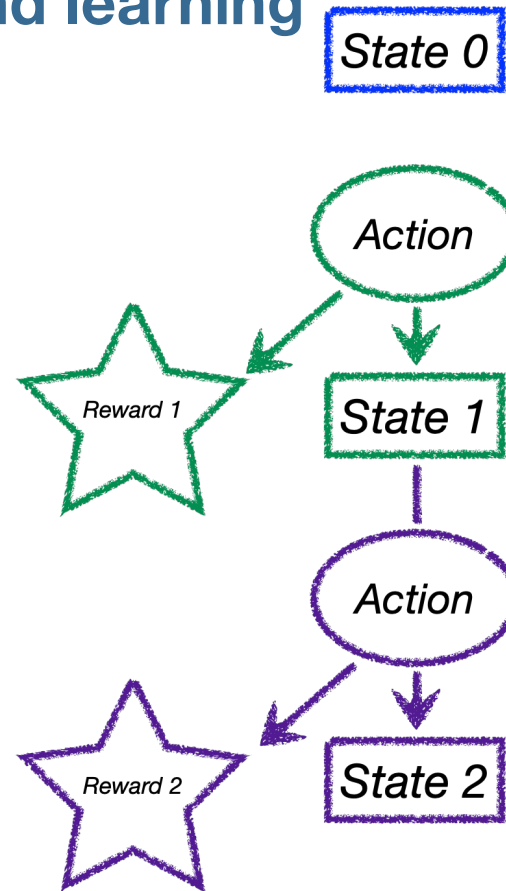
$$(1) \text{ Prob}\{X_{t+T} = X | X_t = x_0, X_{t+1} = x_1, \dots\} = P(X, x_0, x_1, \dots; T)$$

These P 's are assumed not to depend on, or change with, historical calendar time. In that sense, I posit a "stationary" process (but one consistent with, say, chronic inflation).

Samuelson, *Industrial Management Review*, 1965

Rewards drive actions and learning

- Rewards and punishments, received as **feedback** following the implementation of each action, form the key information set in determining policy
- Evolution may be **stochastic**
 - Randomness in environment
 - Randomness in policy
- Example:
 - 1-period return on security holding
 - Penalty for breaching risk limit



Target objective is to maximize return

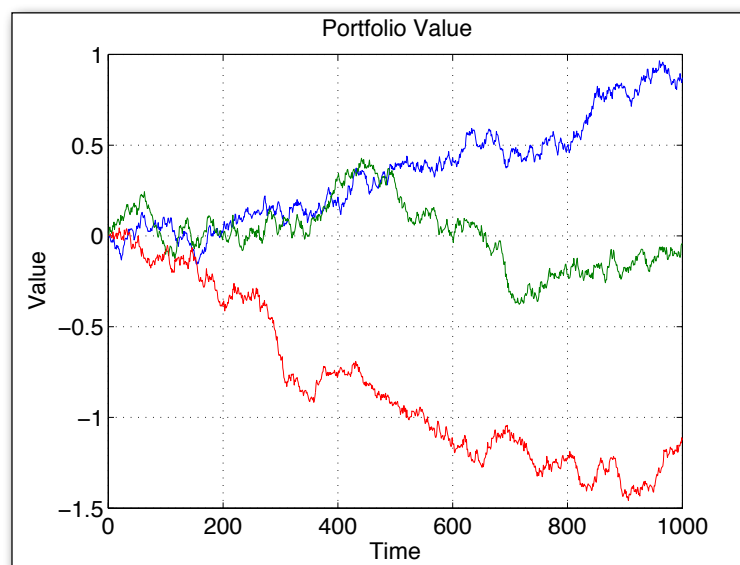
- The essence of reinforcement learning is that we provide only the **final goal**, not the means to achieve it

- Example:

- Present value of future rewards
- Long-term expected return

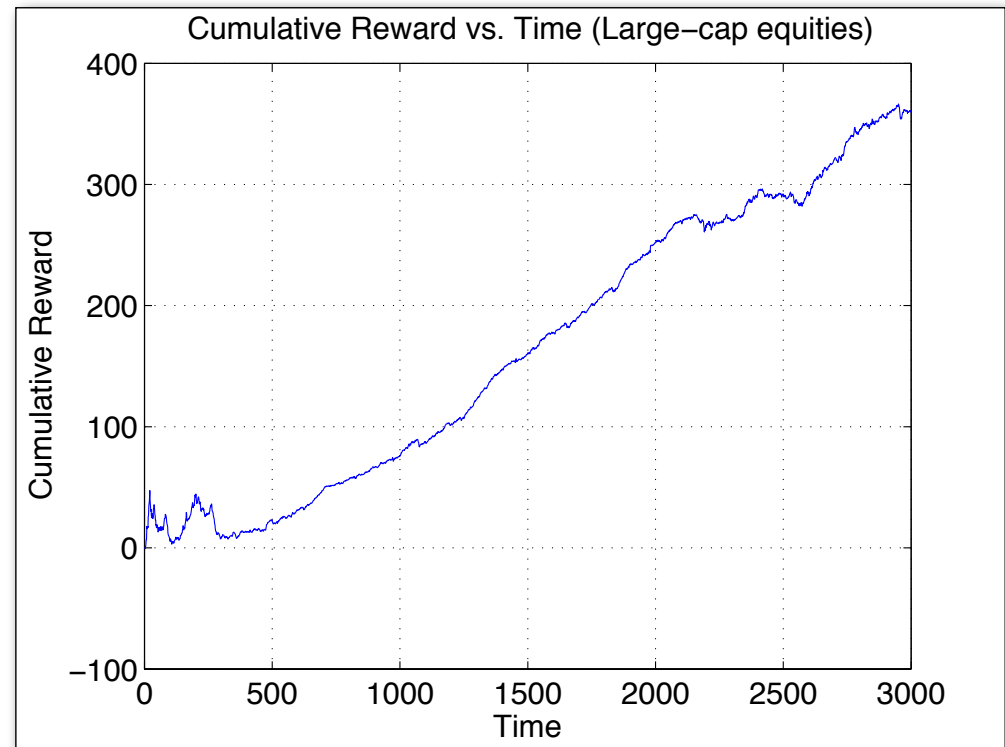
$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

- Long-term Sharpe ratio
- For finite horizon, completion reward



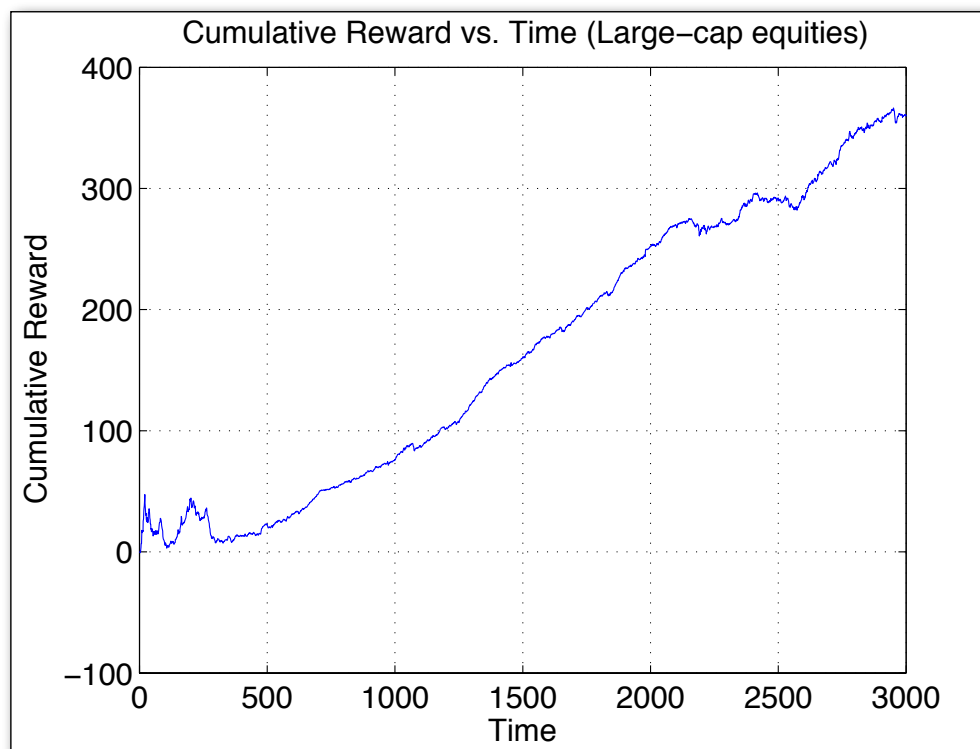
Learning to trade

- Universe:
 - Large cap US equities
 - 600 assets
 - 1-day returns
- States and features:
 - Prior day excess return decile
- Actions:
 - Long, short, flat



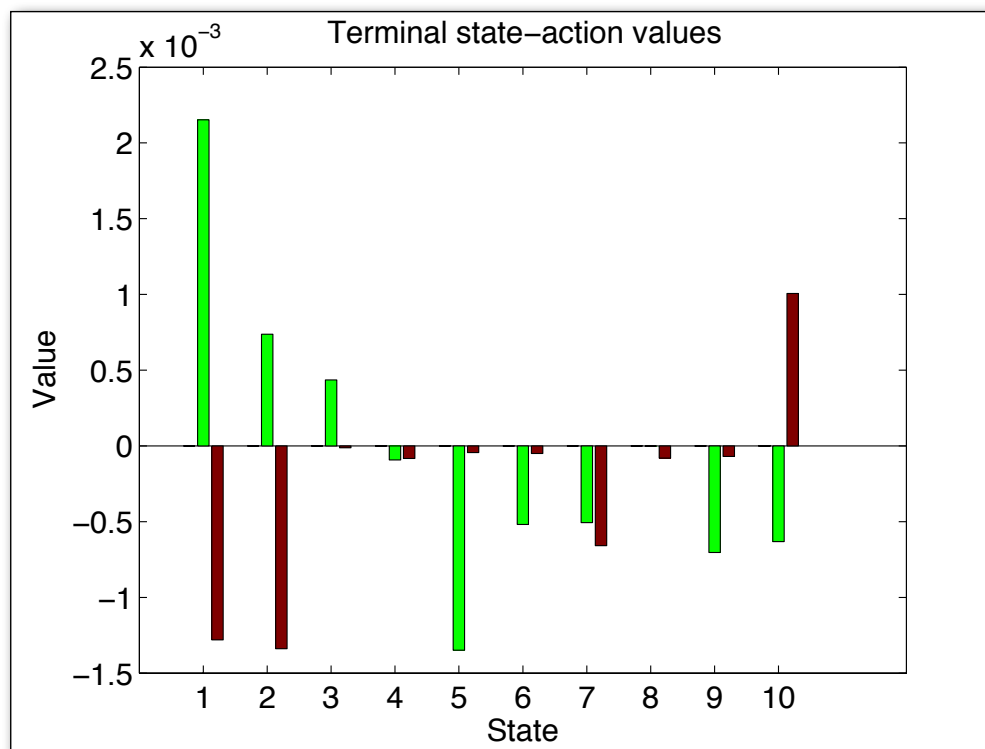
Learning to trade

- Initial learning period
- Steady state
- Market change?
- How does it work??



From “black box” to glass box

- Learned strategy is **interpretable** and **transparent**:
 - Examine the learned state-action values
 - 10 states – worst (1) to best (10)
 - 3 actions – long (green), short (red), or neutral
- Conclusion:
 - ***Buy the losers***
 - ***Sell the winners***



Market behavior and behavior in markets

When Are Contrarian Profits Due to Stock Market Overreaction?

Andrew W. Lo
Sloan School of Management
Massachusetts Institute of Technology

A. Craig MacKinlay
Wharton School
University of Pennsylvania

If returns on some stocks systematically lead or lag those of others, a portfolio strategy that sells "winners" and buys "losers" can produce positive expected returns, even if no stock's returns are negatively autocorrelated as virtually all models of overreaction imply. Using a particular contrarian strategy we show that, despite negative autocorrelation in individual stock returns, weekly portfolio returns are strongly positively autocorrelated and are the result of important cross-autocorrelations. We find that the returns of large stocks lead those of smaller stocks, and we present evidence against overreaction as the only source of contrarian profits.

Since the publication of Louis Bachelier's thesis *Theory of Speculation* in 1900, the theoretical and empir-

The Review of Financial Studies, v 3 n 2 1990

ical implications of the random walk hypothesis as a model for speculative prices have been subjects of considerable interest to financial economists. First developed by Bachelier from rudimentary economic considerations of "fair games," the random walk has received broader support from the many early empirical studies confirming the unpredictability of stock-price changes.¹ Of course, as Leroy (1973) and Lucas (1978) have shown, the unforecastability of asset returns is neither a necessary nor a sufficient condition of economic equilibrium. And, in view of the empirical evidence in Lo and MacKinlay (1988), it is also apparent that historical stock market prices do not follow random walks.

This fact surprises many economists because the defining property of the random walk is the uncorrelatedness of its increments, and deviations from this hypothesis necessarily imply price changes that are forecastable to some degree. But our surprise must be tempered by the observation that forecasts of stock returns are still imperfect and may be subject to considerable forecast errors, so that "excess" profit opportunities and market inefficiencies are not necessarily consequences of forecastability. Nevertheless, several recent studies maintain the possibility of significant profits and market inefficiencies, even after controlling for risk in one way or another.

Some of these studies have attributed this forecastability to what has come to be known as the "stock market overreaction" hypothesis, the notion that investors are subject to waves of optimism and pessimism and therefore create a kind of "momentum" that causes prices to temporarily swing away from their fundamental values [see, e.g., DeBondt and Thaler (1985, 1987), DeLong, Shleifer, Summers, and Waldmann (1989), Lehmann (1988), Poterba and Summers (1988), and Shefrin and Statman (1985)]. Although such a hypothesis does imply predictability, since what goes down must come up and vice versa, a well-articulated equilibrium theory of overreaction with sharp empirical implications has yet to be developed.

But common to virtually all existing theories of overreaction is one very specific empirical implication: Price changes must be *negatively* autocorrelated for some holding period. For example, DeBondt and

Lo & MacKinlay, *Review of Financial Studies*, 1990

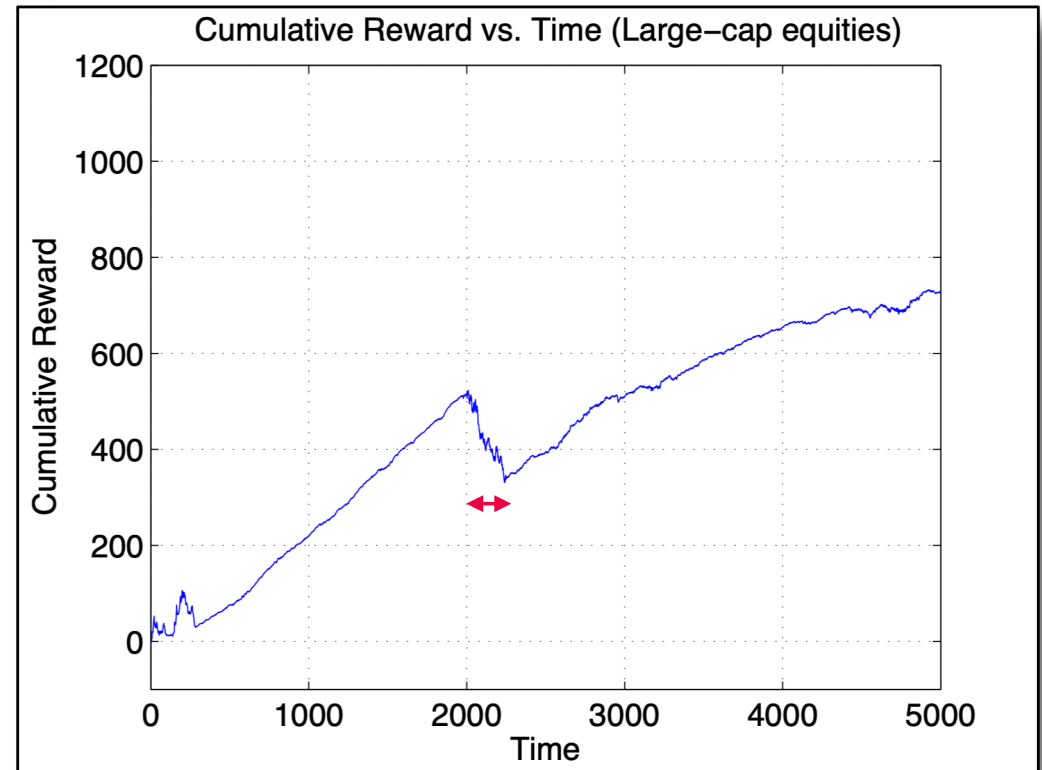
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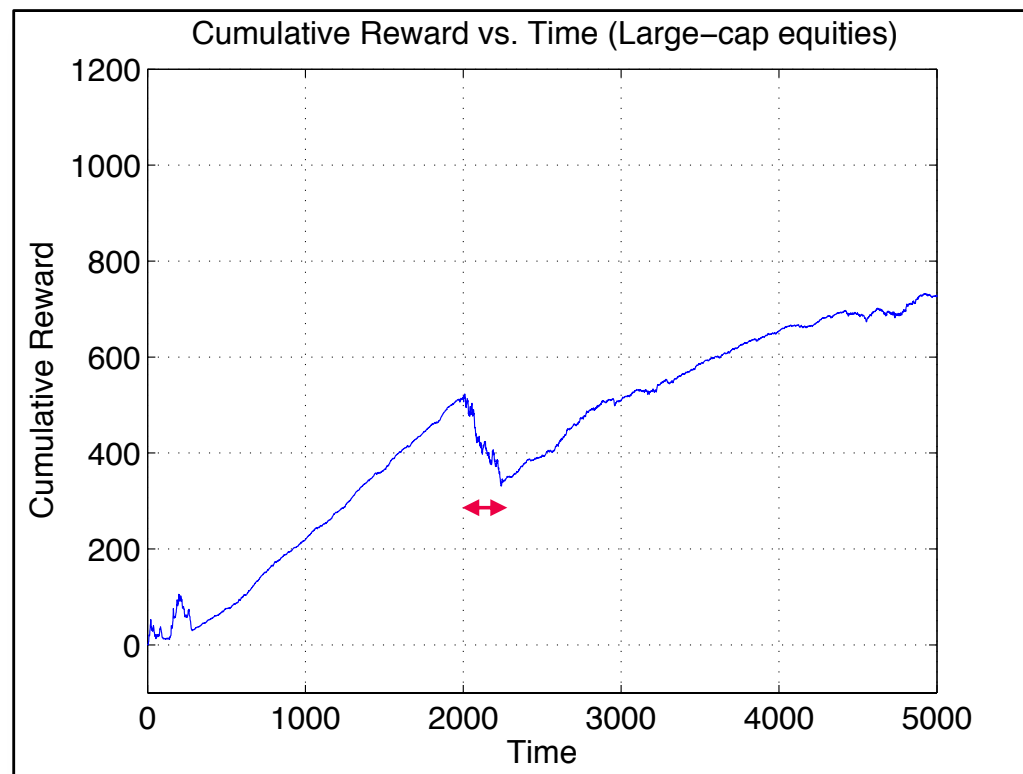
Markets change. Should the model change too?

- What happens if signals **change**?
 - Financial markets are **non-stationary**.
 - How **should** a predictive model behave when the rules of the game are changed?
- Example: suppose market prices become **truly random**?



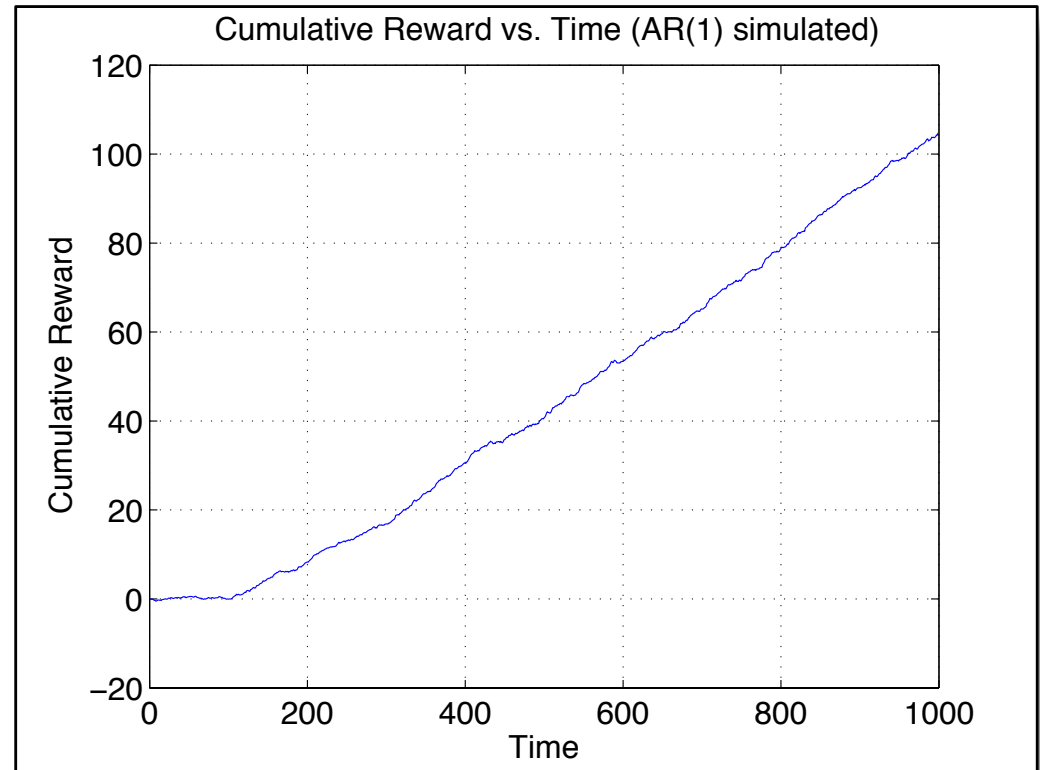
Learning to lose

- Simulation:
 - Market data replaced by **random returns**
- How **should** a predictive model behave when the rules of the game are changed?

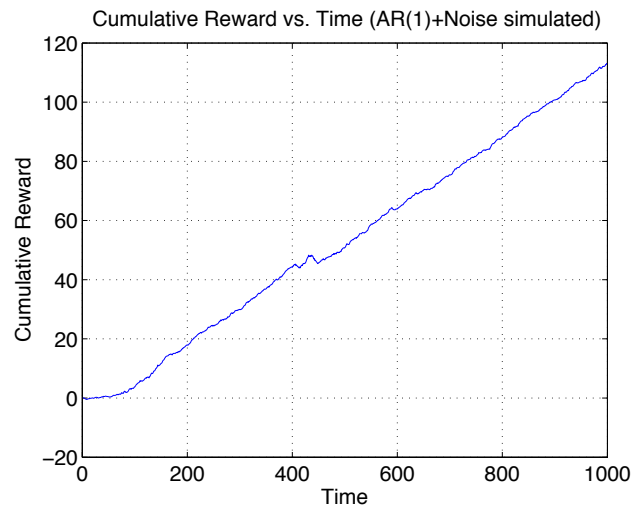
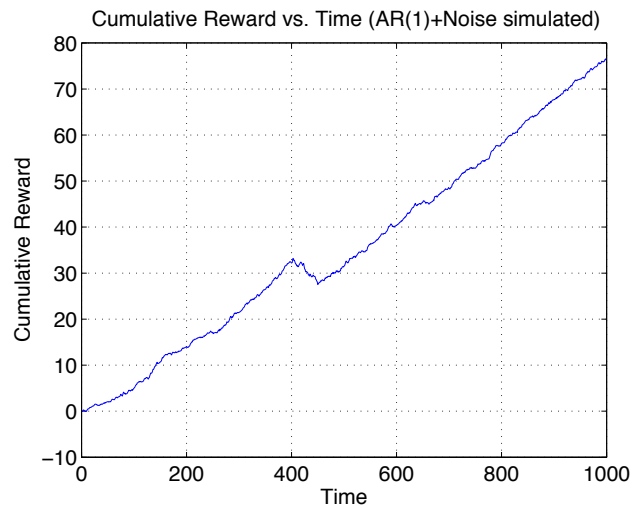


Generative testing

- Run algorithms against **known signals**
 - Generate synthetic data from a **known** probability distribution
 - Remove historical trends and accidents
 - Study evolution in homogeneous environment
 - Develop robust and computationally efficient techniques
 - Provide a baseline for testing against real data

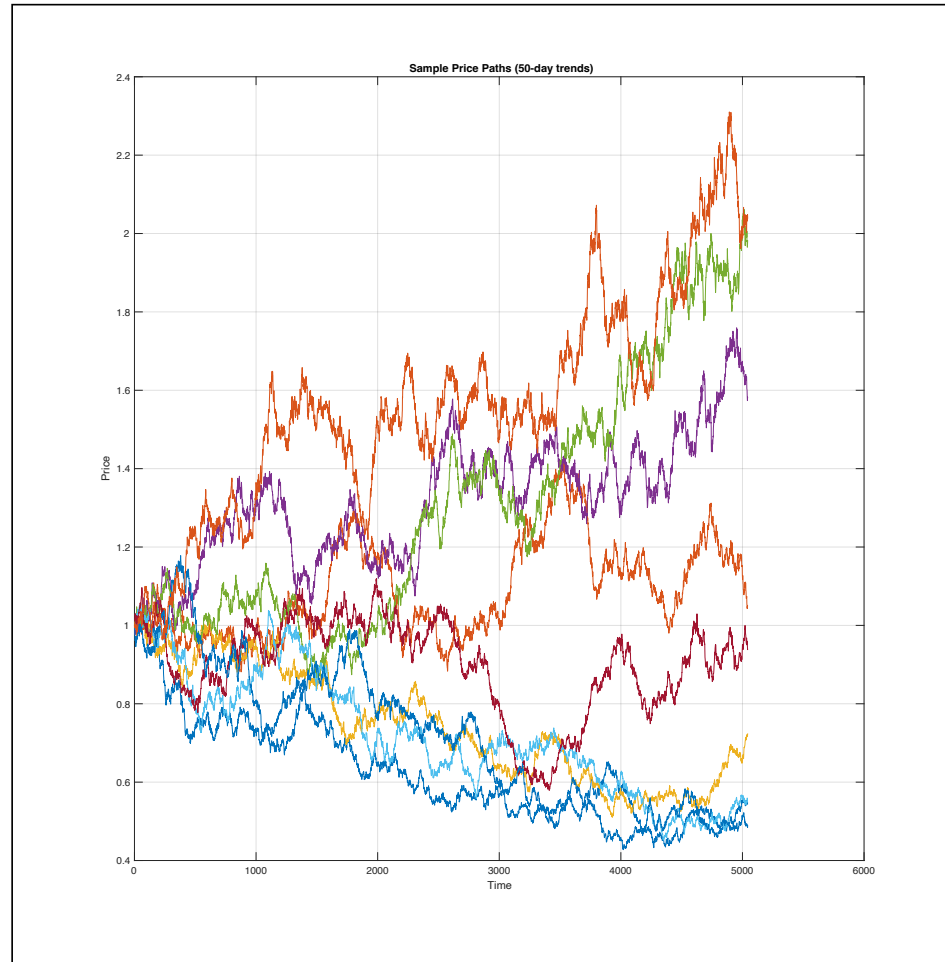


Probing regime change and adaptation



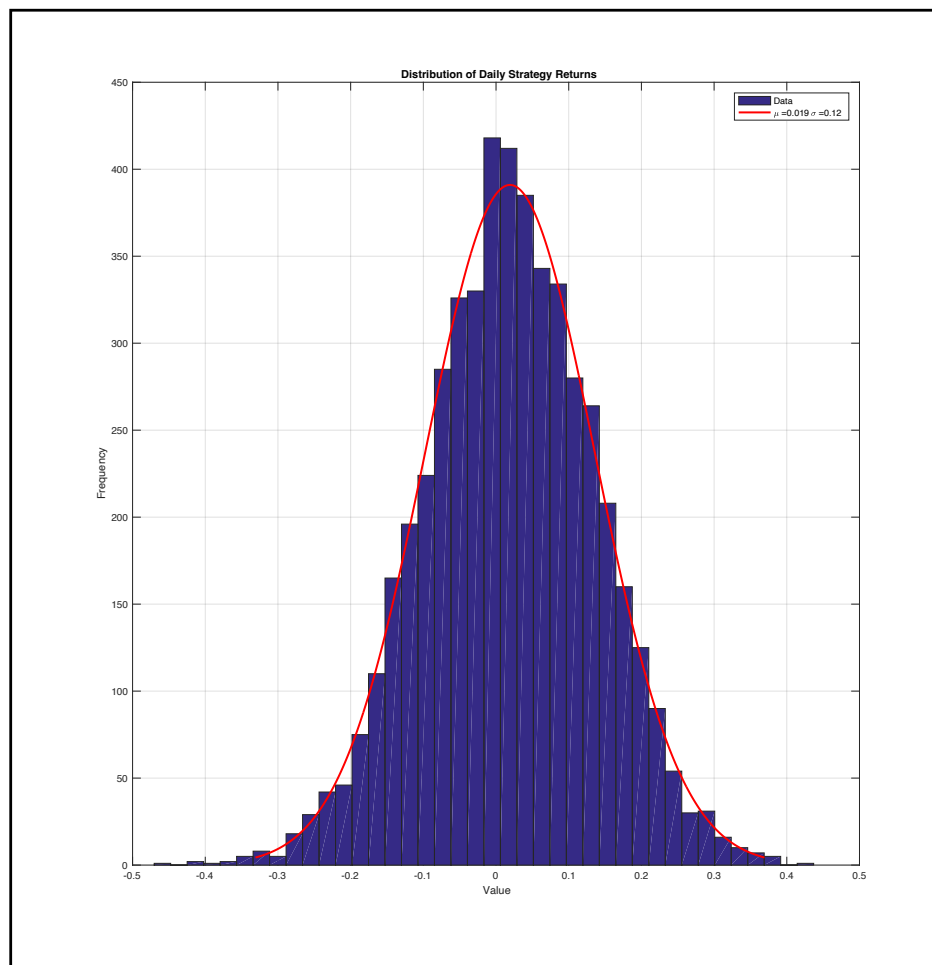
Beyond mean reversion

- Can RL detect **other** kinds of signals?
- Can RL detect **changes**?
 - Market environments evolve
 - One signal decays
 - New signal emerges
 - Regime change
- **Experiment:**
 - New data set, new (simulated) price paths
 - Statistically, looks like GBM
 - Pictorially, too



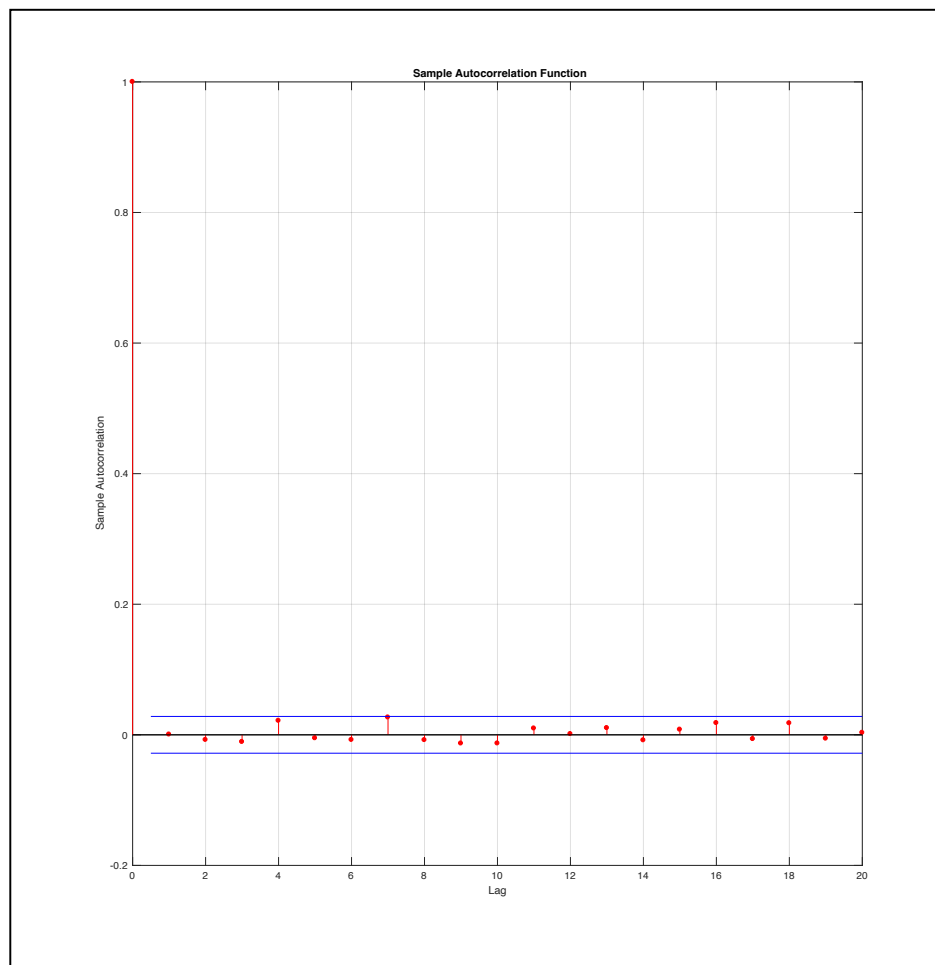
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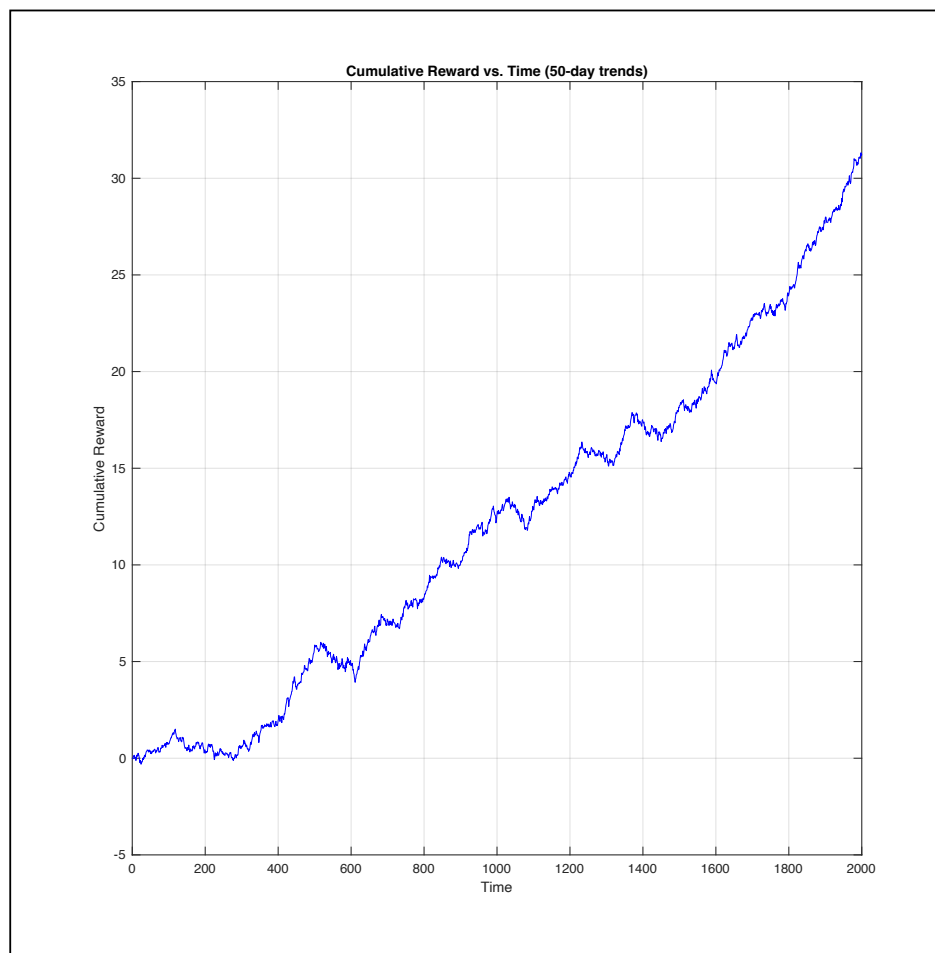
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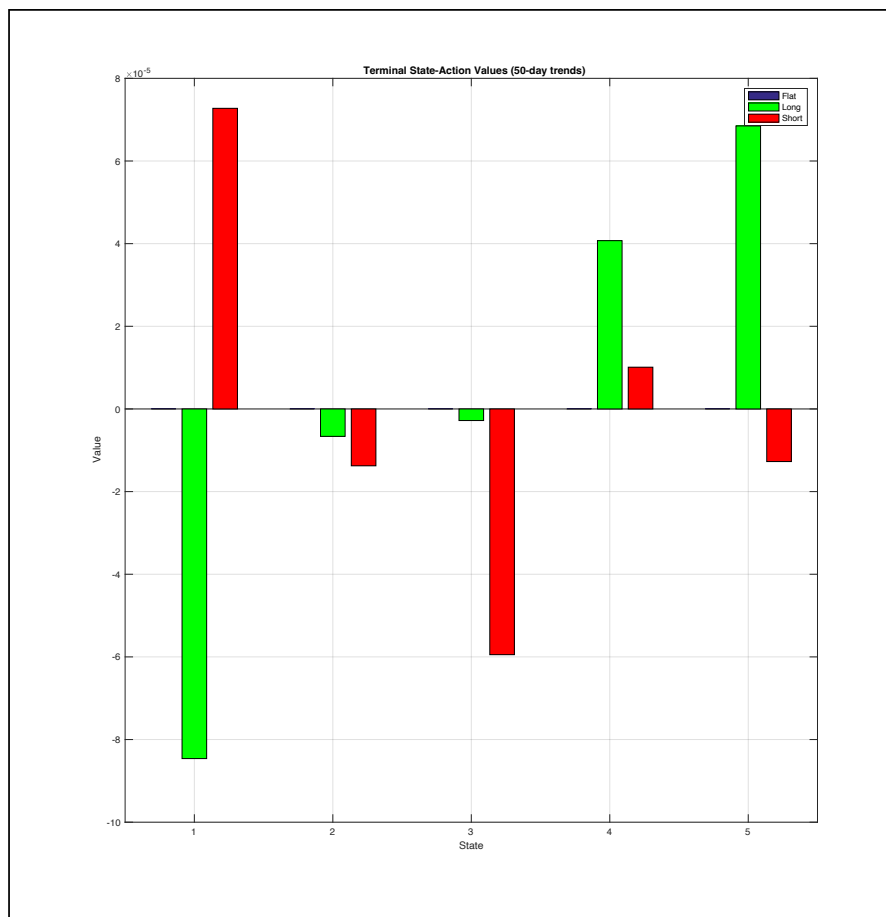
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- **Experiment:**
 - New data set, new (simulated) price paths
 - Statistically, looks like GBM
 - Pictorially, too
 - Data contains randomly distributed **trends**



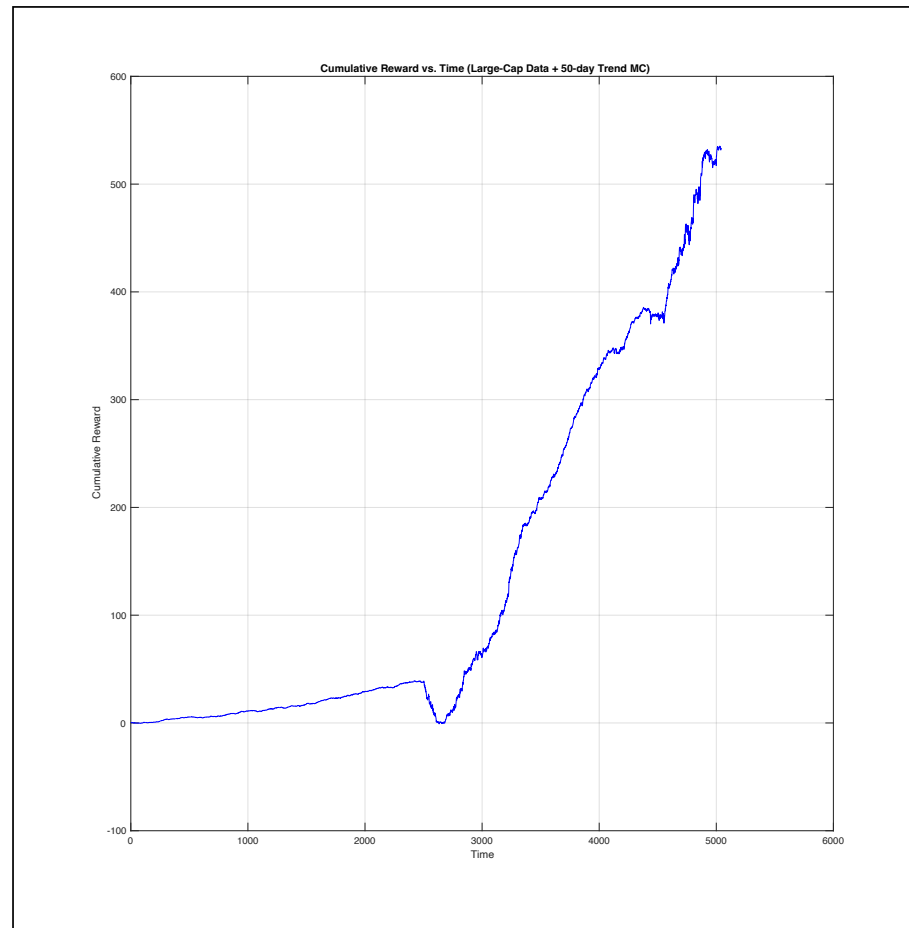
Inside the black box

- Determination and evolution of strategy can be followed from history of internal state variables
- Interpretability:
 - Examine the learned state-action values
 - 5 states – worst (1) to best (5)
 - 3 actions – long (green), short (red), or neutral
- Conclusion:
 - ***Buy the winners***
 - ***Sell the losers***



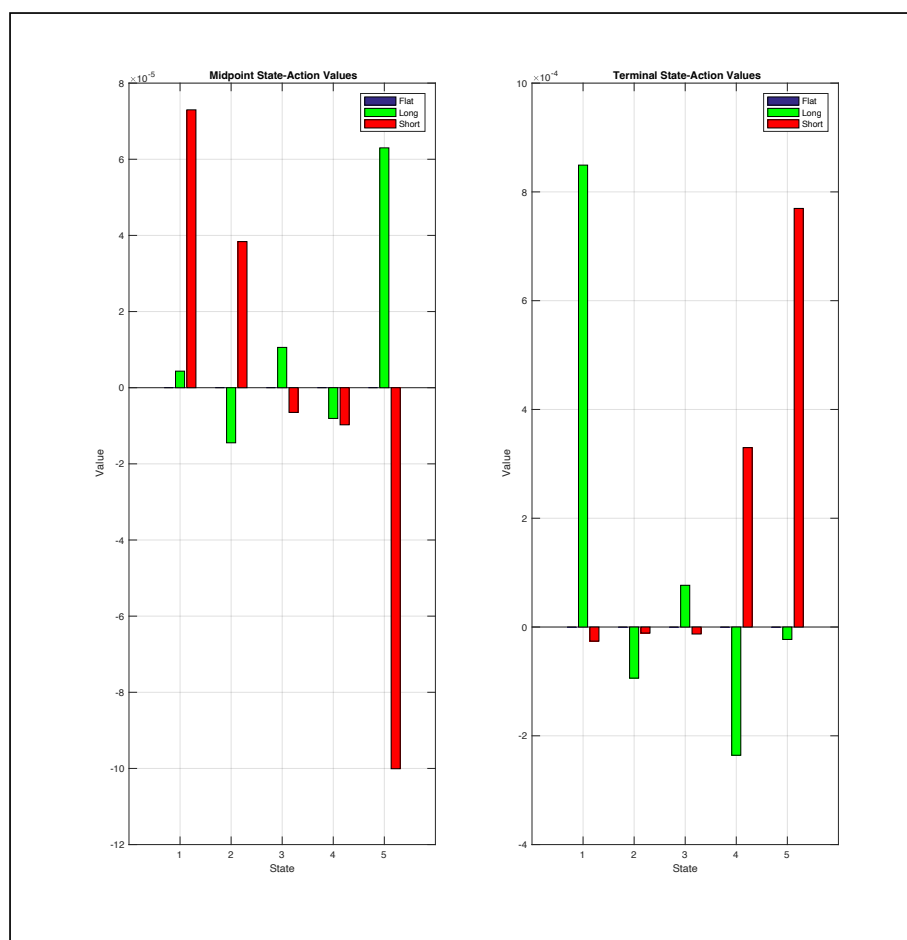
Learning from failure

- How do strategies fail?
- How do we know they have failed?
- What do we do next?
- Experiment:
 - Simulated regime shift
 - MC: 10 years of trend-following
 - CRSP: 10 years of market returns



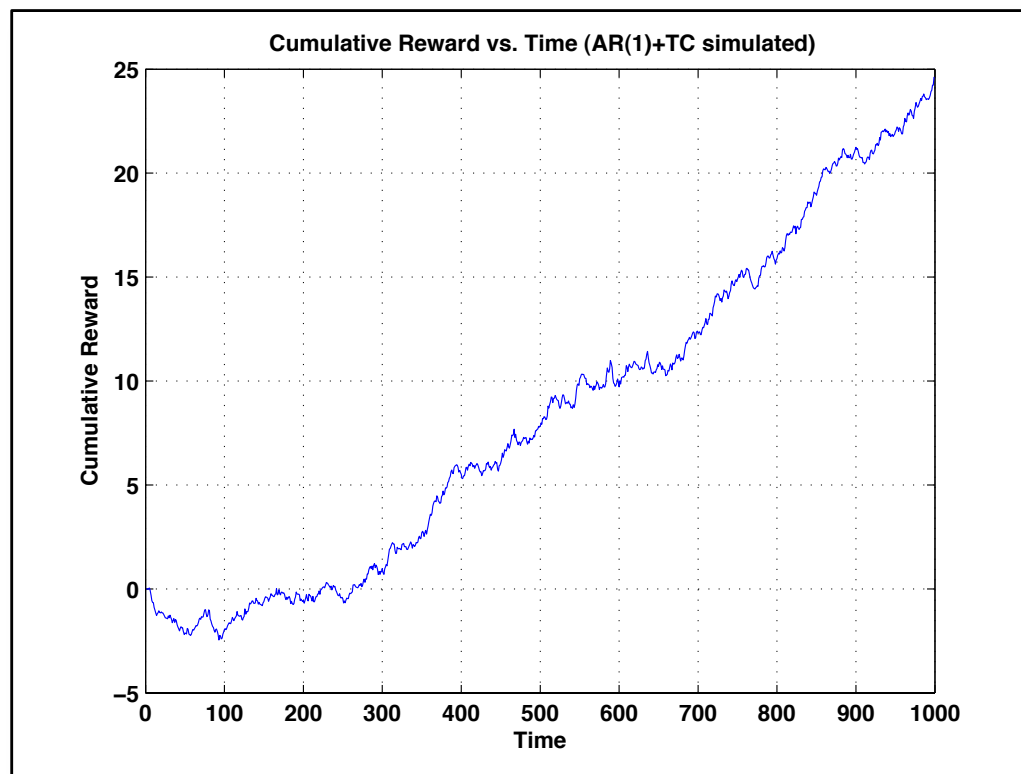
Inside the black box

- Provisional beliefs
 - **Midpoint** state-action values vs.
 - **Endpoint** state-action values
- Evolution and transition
- Speed and success of adaptation depend on
 - Exploration budget
 - Signal strength
 - Availability of relevant variables

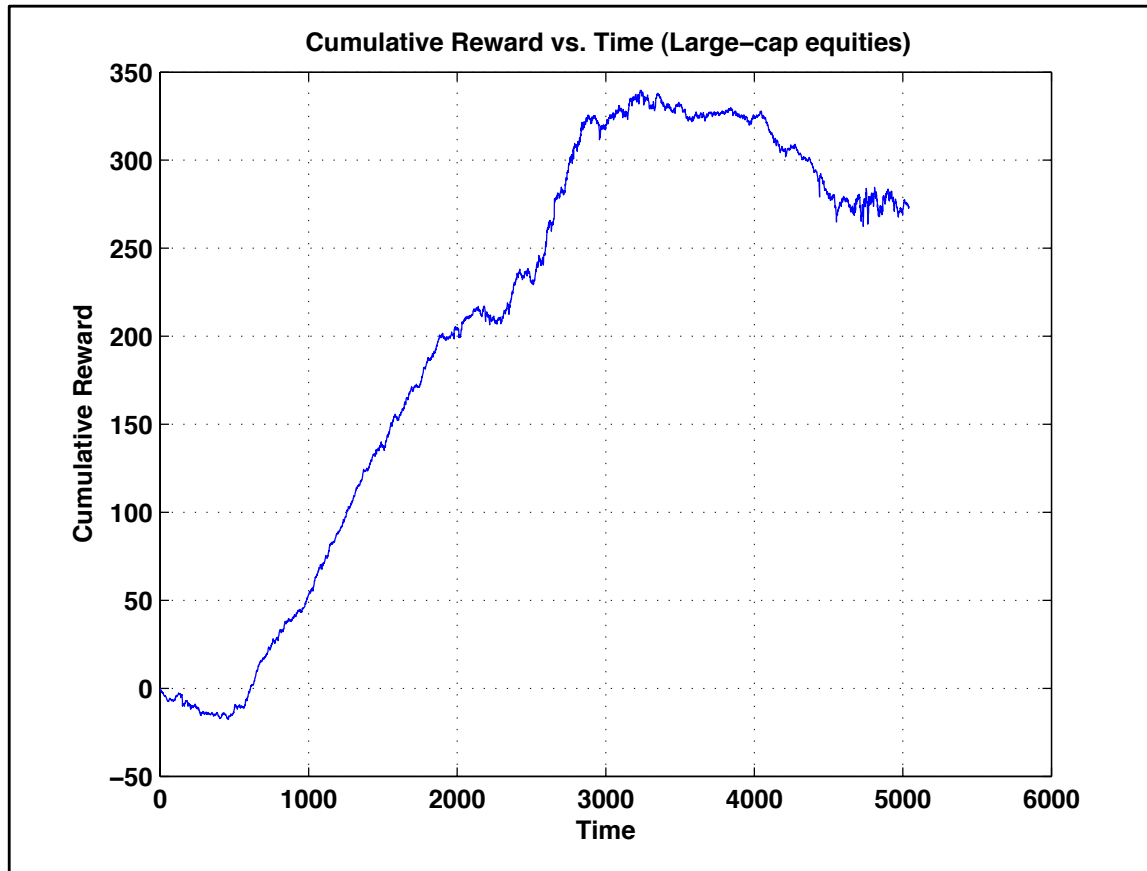


Complete the model

- Larger state space
 - Include portfolio holdings
- Temporal dependence and credit assignment
- **Transaction costs**
- Consequences
 - Opportunity costs
 - Buy vs. hold
 - Non-trading



Reality check: financial frictions in action



RL takeaways

- Reinforcement learning is a technique that is particularly well suited to financial problems where
 - "True" rules are unknown
 - Data is noisy
 - Information acquisition is costly
 - Markets are changing and adaptive
 - Interpretability matters
- Laboratory for
 - **learning to trade**, identifying features and their values
 - **trading to learn** about market dynamics
- Balances exploration, exploitation, and learning rate
- Discovers best actions to achieve goals through trial, error, and learning

Good luck!
Q&A



Finance at MIT
Where ingenuity drives results